

Smart Clinic Management System

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Overview

This project is a comprehensive and scalable Management System designed to streamline clinic operations and enhance patient care. It provides a robust set of features for managing patient records, appointments, medical prescriptions, and invoicing, serving as a powerful platform for healthcare providers to manage their daily workflows efficiently.

Features

- **User Authentication & Authorization:** Secure user registration, login, and role-based access control (Admin, Doctor, Receptionist) using ASP.NET Core Identity.
- **Patient Management:** Comprehensive CRUD operations for patient profiles, including personal details and medical history.
- **Appointment Scheduling:** Efficiently manage patient appointments with search, create, update, and status tracking functionalities.
- **Medical Prescriptions:** Generate and manage digital prescriptions linked to a comprehensive medicine database.
- **Invoice & Billing:** Create and manage invoices for medical services and prescriptions, facilitating streamlined financial tracking.
- **Visit Tracking:** Document and track every patient visit, ensuring a detailed chronological medical record.
- **Department & Doctor Management:** Organize clinic structure by departments and manage professional profiles for medical staff.
- **Dashboard Analytics:** Interactive dashboard providing a high-level overview of clinic activities and key metrics.

Architecture & Design Principles

This system is built with a focus on maintainability, scalability, and performance, adhering to modern software engineering practices:

- **MVC Pattern:** Implemented using the Model-View-Controller architecture to ensure a clean separation of concerns.
- **Repository Pattern:** Utilized to abstract data access logic, ensuring a clean separation between business logic and data persistence. This enhances testability and allows for easier database changes.
- **Dependency Injection:** Extensively used to manage service dependencies, promoting modularity and flexibility across the application.
- **View Models (VMs):** Used to shape and validate data sent to and received from views,

protecting internal domain models and optimizing data transfer.

- **Identity Security:** Provides a secure, cookie-based authentication mechanism for managing user sessions and roles.
- **Error Handling:** Centralized exception handling to provide consistent and meaningful error responses throughout the application.

Technologies Used

- **Backend:**
 - C#
 - ASP.NET Core MVC
 - Entity Framework Core (ORM)
 - SQL Server (Database)
 - LINQ (Language Integrated Query)
 - ASP.NET Core Identity (User Management)
 - AutoMapper (Object-to-Object Mapping)
- **Frontend:**
 - HTML5 & CSS3
 - JavaScript & jQuery
 - Bootstrap (Responsive UI Framework)

Getting Started

To get the Smart Clinic Management System up and running on your local machine for development and testing purposes, follow these steps:

Prerequisites

- .NET SDK (version 6.0 or higher)
- SQL Server (or SQL Server Express)
- Visual Studio (or Visual Studio Code with C# extension)

Installation

1. **Clone the repository:** `bash git clone https://github.com/YOUR_USERNAME/SmartClinic.git cd SmartClinic`
2. **Configure Database:**
 - Open `appsettings.json` in the `smart clinic` project.
 - Update the `DefaultConnection` connection string to point to your local SQL Server instance.
 - Run database migrations: `bash dotnet ef database update`
3. **Run the application:** `bash dotnet run` The application will typically run on `https://localhost:5001` (or a similar port).

Contributing

Feel free to fork the repository, create a new branch, and submit pull requests. Any contributions are welcome!

License

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Contact

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